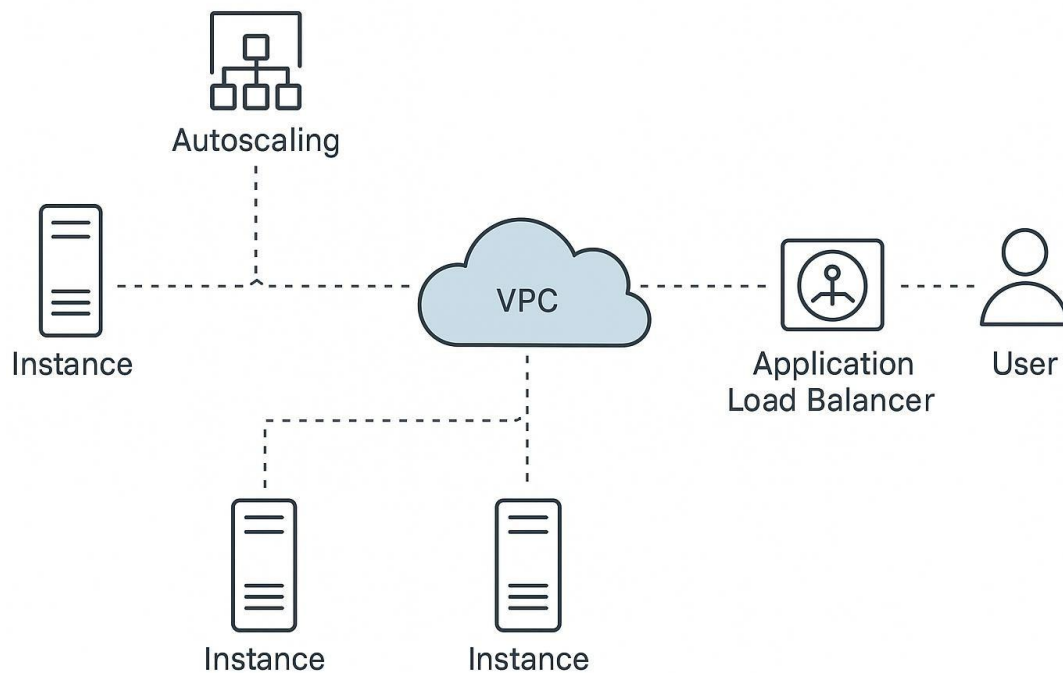


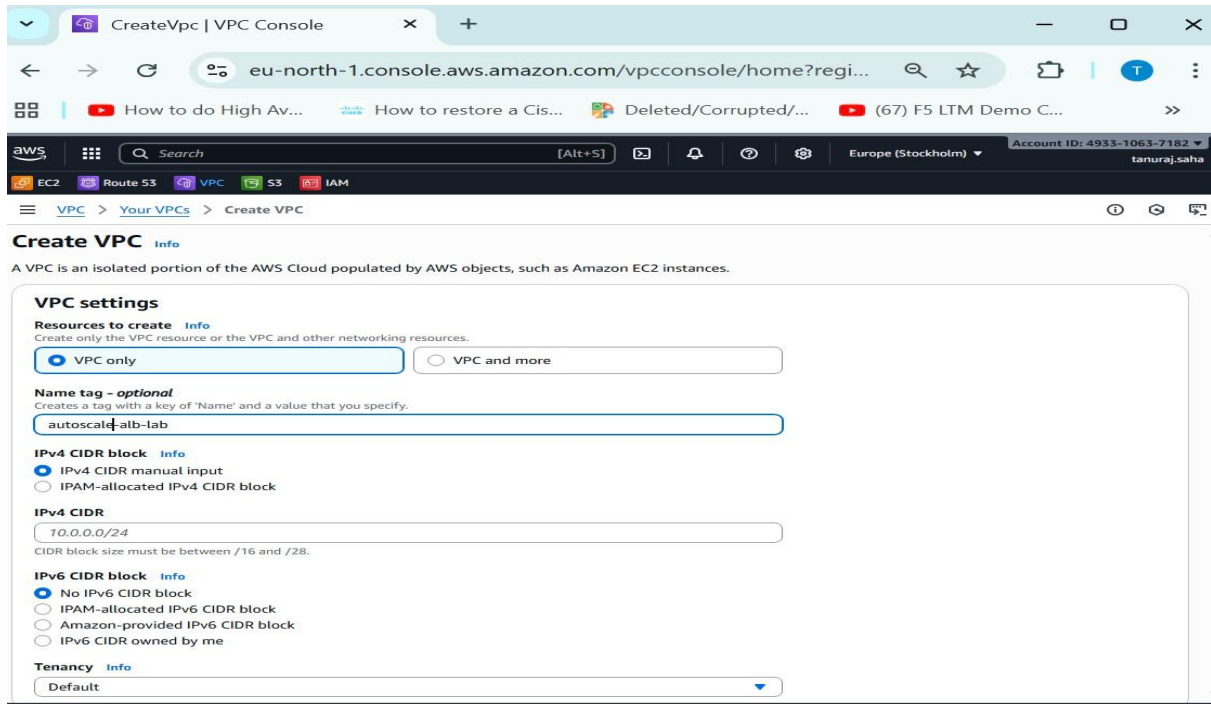
AWS Auto Scaling Group with Load Balancer

Summary: Setting up an Autoscaling Group after an Application Load Balancer in AWS to manage traffic and dynamically adjust capacity. When traffic from outside hits the ALB it will automatically load-balance to the multiple servers instead of one to ensure the load sharing. Also, these servers are created using template and auto-scaling group, so we don't need manual provisioning all the time.

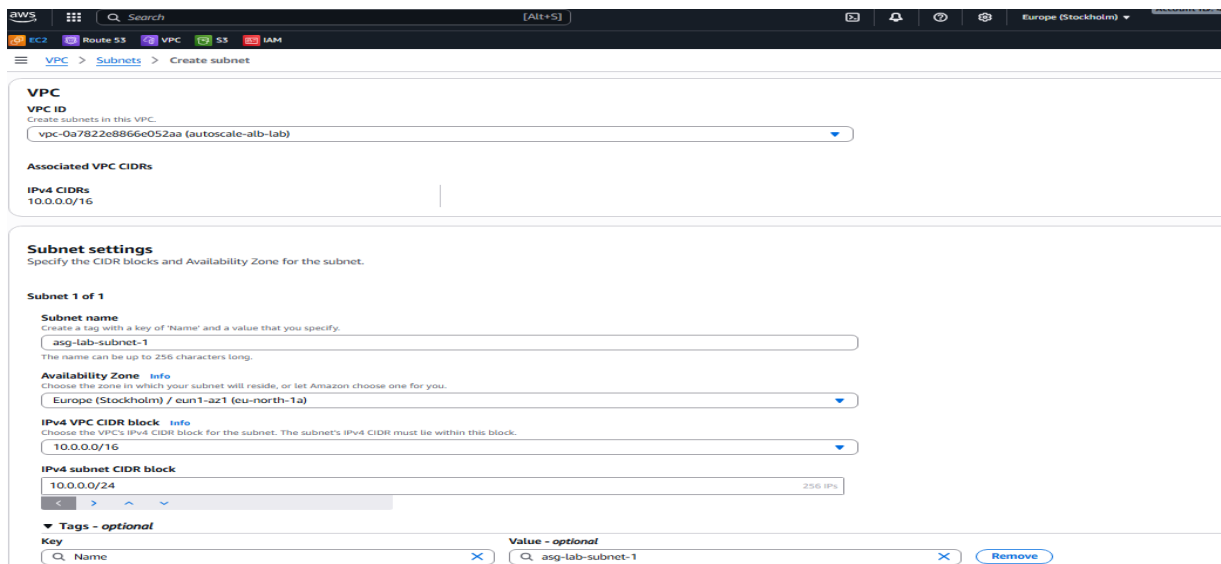
Here is the lab topology below:



Step-1: Create a new VPC - AWS Console → VPC → Create VPC - Name : autoscale-alb-lab, IPv4 CIDR block: 10.0.0.0/16 - Click “Create VPC”



Step-2: Create Subnet - VPC → Subnets → Create Subnet - VPC ID: Select autoscale-alb-lab - Subnet name: asg-lab-subnet-1 - Availability Zone: Select one (Europe-Stockholm / eu-north-1a) - CIDR block: 10.0.1.0/24 - Auto-assign public IPv4 address: Enable for public subnet - Click “Create Subnet”



Do the same for another subnet-2 (10.0.2.0/24)

AWS [Search] [Alt+S] Europe (Stockholm) Account ID: 4933-

EC2 Route 53 VPC S3 IAM

VPC > Subnets > Create subnet

VPC ID
Create subnets in this VPC.
vpc-0a7822e8866e052aa (autoscale-alb-lab)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
asg-lab-subnet-2
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Europe (Stockholm) / eu-north-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Name	asg-lab-subnet-2	Remove

Step-3: Create and Attach Internet Gateway - VPC → Internet Gateways → Create Internet Gateway - Name : asg-igw - Click “Create Internet Gateway” - Select newly created IGW → Actions → Attach to VPC → autoscale-alb-lab

EC2 Route 53 VPC S3 IAM

VPC > Internet gateways > Create internet gateway

Create internet gateway [Info](#)
An internet gateway is a virtual internet router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Create a tag with a key of 'Name' and a value that you specify.
asg-igw

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	asg-igw	Remove

[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create internet gateway](#)

EC2 Route 53 VPC S3 IAM

VPC > Internet gateways > Attach to VPC (igw-03d876e8d27b20f18)

Attach to VPC (igw-03d876e8d27b20f18) [Info](#)
The following internet gateway was created: igw-03d876e8d27b20f18 - asg-igw. You can now attach to a VPC to enable the VPC to communicate with the internet. [Attach to a VPC](#)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.
vpc-0a7822e8866e052aa

AWS Command Line Interface command

[Cancel](#) [Attach internet gateway](#)

Step-4: Configure Route Table - VPC → Route Tables → Create route table - Name: asg-rt-alb VPC: autoscale-alb-lab - Open Subnet Associations → Edit → Select asg-rt-alb→ Edit routes → Add: - Destination: 0.0.0.0/0 - Target: asg-igw - Save

Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
asg-rt-alb

VPC
The VPC to use for this route table.
vpc-0a7822e8866e052aa (autoscale-alb-lab)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q asg-rt-alb X Remove

Add new tag
You can add 49 more tags.

Cancel Create route table

Route tables (1/3) info

Find route tables by attribute or tag

	Name	Route...	Explicit subnet associations	Edge...	Main	VPC
☐	asg-rt-alb	rtb-0b3...	-	-	No	vpc-0a7822e8866e052aa autoscale-alb-lab
☒	-	rtb-006...	2 subnets	-	Yes	vpc-0a7822e8866e052aa autoscale-alb-lab
☐	-	rtb-000...	-	-	Yes	vpc-0c80a831b80dada5b

Create default route towards 'asg-igw'

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Add route Cancel Preview Save changes

Associate 2 created subnets to the main route-table.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/2)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	asg-lab-subnet-1	subnet-00fe8d03cfa20c1f0	10.0.1.0/24	-	Main (rtb-00669204d33580d5c)
☒	asg-lab-subnet-2	subnet-0960c2bf2612fdef2	10.0.2.0/24	-	Main (rtb-00669204d33580d5c)

Selected subnets

subnet-00fe8d03cfa20c1f0 / asg-lab-subnet-1 X subnet-0960c2bf2612fdef2 / asg-lab-subnet-2 X

[Cancel](#) [Save associations](#)

Step-5: Create Security Group - EC2 → Security Groups → Create Security Group - Name: asg-alb-sg Description: Allow HTTP/HTTPS and SSH - Inbound rules: - HTTP from 0.0.0.0/0 - HTTPS from 0.0.0.0/0 - SSH from your IP - Outbound: Allow all

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name info
asg-alb-sg
Name cannot be edited after creation.

Description info
Allows SSH access to developers

VPC info
vpc-0a7822e8866e052aa (autoscale-alb-lab)

Inbound rules info

Type info: HTTP
Protocol info: TCP
Port range info: 80
Source info: Anyw...
Description - optional info: 0.0.0.0/0 X
[Delete](#)

[Add rule](#)

Outbound rules info

Type info: All traffic
Protocol info: All
Port range info: All
Destination info: Custom
Description - optional info: 0.0.0.0/0 X
[Delete](#)

[Add rule](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

Step-6: Create Launch Template - EC2 → Launch Templates → Create launch template - Name: asg-alb-template - AMI: Amazon Linux 2 - Instance type: t2.micro - Key pair: Choose existing or create - Security group: asg-alb-sg - User data: (Optional script) - Click “Create launch template”

Launch template name - *required*
asg-alb-template
Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description
web server
Max 255 chars

Auto Scaling guidance [Info](#)
Select this if you intend to use this template with EC2 Auto Scaling
☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags
► Source template

Launch template contents
Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) [Info](#)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents [Quick Start](#)

Don't include in launch template

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux D

Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.1 AMI
ami-0854d4f8e4bd6b834 (64-bit (x86), uefi-preferred) / ami-0a6a8a732ec383681 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs Free tier eligible

Instance type
t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour
On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand SUSE base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour
Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name
asg-alb
Create new key pair

▼ Network settings [Info](#)
Subnet [Info](#)
subnet-00fe8d03cfa20c1f0 Owner: 493310637182
VPC: vpc-0a7822e886e052aa Availability Zone: eu-north-1a (eu1-a21) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.1.0/24
When you specify a subnet, a network interface is automatically added to your template.
Availability Zone [Info](#)
eu-north-1a eu1-a21
Enable additional zones
Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Select existing security group Create security group
Common security groups [Info](#)
Select security groups
asg-alb-sg sg-099341da1d64e7665
VPC: vpc-0a7822e886e052aa
Security groups that you add or remove here will be added to or removed from all your network interfaces.
► Advanced network configuration

Summary
Software Image (AMI)
Amazon Linux 2023.9.2...[read more](#)
ami-0854d4f8e4bd6b834
Virtual server type (instance type)
t3.micro
Firewall (security group)
asg-alb-sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel Create launch template

aws

Search

[Alt+S]

EC2Route 53VPCs3IAM

Europe (Stockholm)

Account ID: 4933-1063-7182

tanuraj.sah

Create launch template

Metadata IPv6 endpointInfo

Don't include in launch template

Metadata versionInfo

V2 only (token required)

⚠ For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limitInfo

2

Allow tags in metadataInfo

Don't include in launch template

User data - optionalInfo

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "Welcome To Autoscaling lab" > /var/www/html/index.html
```

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-0854d4f8e4bd6b834

Virtual server type (instance type)
t3.micro

Firewall (security group)
asg-alb-sg

Storage (volumes)
1 volume(s) - 8 GiB

CancelCreate launch template

Step-7: Create Application Load Balancer - EC2 → Load Balancers → Create Load Balancer - Choose “Application Load Balancer” - Name: ALB-asg - Scheme: Internet-facing - Listeners: HTTP 80 - Availability Zones: Select VPC and subnet - Security Group: asg-alb-sg



How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
ALB-asg

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **Internet-facing** load balancers only.

Network mapping

[Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0a7822e8866e052aa (autoscale-alb-lab)
10.0.0.0/16

[Create VPC](#)

IP pools | [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools](#) in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Select the previously created vpc (autoscale-alb-lab) and select the both subnets(10.0.1.0/24 and 10.0.2.0/24)

Network mapping | [Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0a7822e8866e052aa (autoscale-alb-lab)
10.0.0.0/16

[Create VPC](#)

IP pools | [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools](#) in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **eu-north-1a (eu-n1-az1)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-00fe8d03cfa20c1f0
IPv4 subnet CIDR: 10.0.1.0/24

asg-lab-subnet-1

☒ **eu-north-1b (eu-n1-az2)**

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0960c2bf2612fdef2
IPv4 subnet CIDR: 10.0.2.0/24

asg-lab-subnet-2

Step-8: Create Target Group - EC2 → Target Groups → Create target group - Target type: Instance - Protocol: HTTP - Port: 80 - VPC: autoscale-alb-lab; VPC - Name: tg-asg-alb Click “Create target group”

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

- ☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB
- ☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB
- ☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB
- ☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

tg-asg-alb

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-52 total characters; Count: 0/52

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

- ☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0a7822e8866e052aa (autoscale-alb-lab)

10.0.0.0/16

[Create VPC](#)

Protocol version

- ☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.
- ☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.
- ☐ **gRPC**
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

After that the available instance needs to be registered like below.

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 4933-1063-7182

tanuraj.saha

EC2Route S3VPCIAM

EC2Target groups>tg-asg-alb>Register targets

Register targets

Select instances, specify ports, and add the instances to the list of pending targets. Repeat to add additional combinations of instances and ports to the list of pending targets. Once you are satisfied with your selections, click Register pending targets.

Available instances (1/1)

< 1 >

☒	Instance ID	Name	State	Security groups	Zone	Private IPv4 address
☒	i-04e85f335d66077dd		Running	asg-alb-sg	eu-north-1a	10.0.1.14

1 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with commas)

Include as pending below

Review targets

Targets (0)

Show only pending

Remove all pending

< 1 >

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
-------------	------	------	-------	-----------------	------	----------------------	-----------	-------------

No instances added yet

Specify instances above, or leave the group empty if you prefer to add targets later.

aws

Search

[Alt+S]

Europe (Stockholm)

Account ID: 4933-1063-7182

tanuraj.saha

EC2Route S3VPCIAM

EC2Target groups>tg-asg-alb>Register targets

Register targets

Select instances, specify ports, and add the instances to the list of pending targets. Repeat to add additional combinations of instances and ports to the list of pending targets. Once you are satisfied with your selections, click Register pending targets.

Available instances (1)

< 1 >

☐	Instance ID	Name	State	Security groups	Zone	Private IPv4 address
☐	i-04e85f335d66077dd		Running	asg-alb-sg	eu-north-1a	10.0.1.14

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Show only pending

Remove all pending

< 1 >

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-04e85f335d66077dd		80	Running	asg-alb-sg	eu-north-1a	10.0.1.14	subnet-00fe8d03cfa20c1f0	October 24, 2025, 13:36 (UTC+03:00)

1 pending

Cancel

Register pending targets

Step-9: Create Auto Scaling Group - EC2 → Auto Scaling Groups → Create Auto Scaling group - Launch Template: - Group name: ASG-lab-group and Subnets: Select previously created - Load Balancing: Attach to ALB-asg and Target Group : tg-asg-lab - Desired Capacity: 2 - Min/Max: 1/3 - Click “Create Auto Scaling Group”

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The breadcrumb navigation indicates the path: EC2 > Auto Scaling groups > Create Auto Scaling group. A progress bar on the left lists seven steps: 1. Choose launch template (active), 2. Choose instance launch options, 3. optional - Integrate with other services, 4. optional - Configure group size and scaling, 5. optional - Add notifications, 6. optional - Add tags, and 7. Review.

Choose launch template Info
Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name
Auto Scaling group name
Enter a name to identify the group.
ASG-lab-group
Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info
For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
asg-alb-template

Version
Default (1)

Description
web server

AMI ID
ami-0854d4f8e4bd6b834

Key pair name
asg-alb

Launch template
asg-alb-template
lt-0d065f00f6a0503d3

Instance type
t3.micro

Security groups
-

Security group IDs
sg-099341da1d64e7665

Request Spot Instances
No

Additional details

Storage (volumes)
-

Date created
Fri Oct 24 2025 12:56:45 GMT+0300 (Eastern

The screenshot shows the 'ASG-lab-group' page in the AWS Management Console. The 'Capacity overview' section displays: Desired capacity: 2, Scaling limits (Min - Max): 1 - 5, Desired capacity type: Units (number of instances), and Status: -. The 'Details' tab is selected, showing sections for Load balancing, VPC Lattice integration options, and Application Recovery Controller (ARC) zonal shift.

Group size ×
Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.

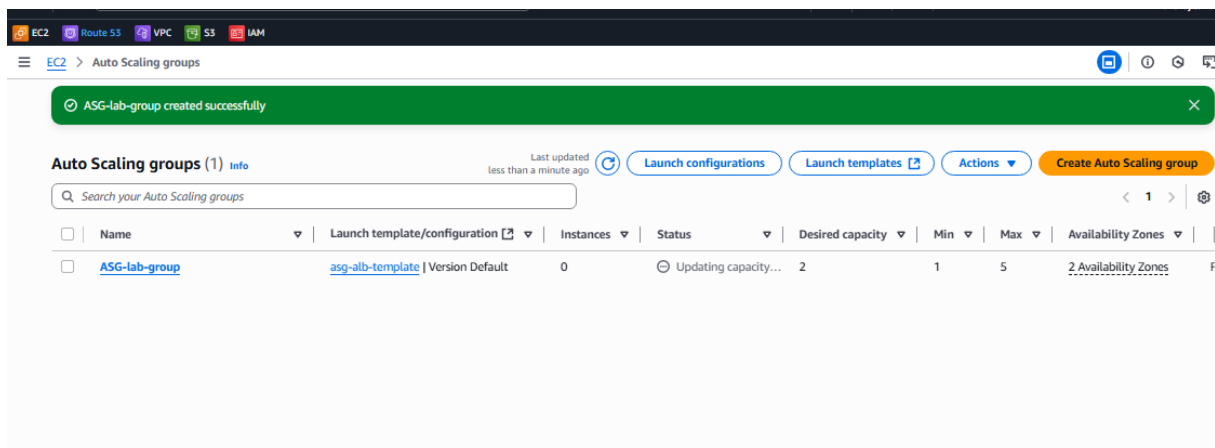
Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.
Units (number of instances)

Desired capacity
Specify your group size.
2

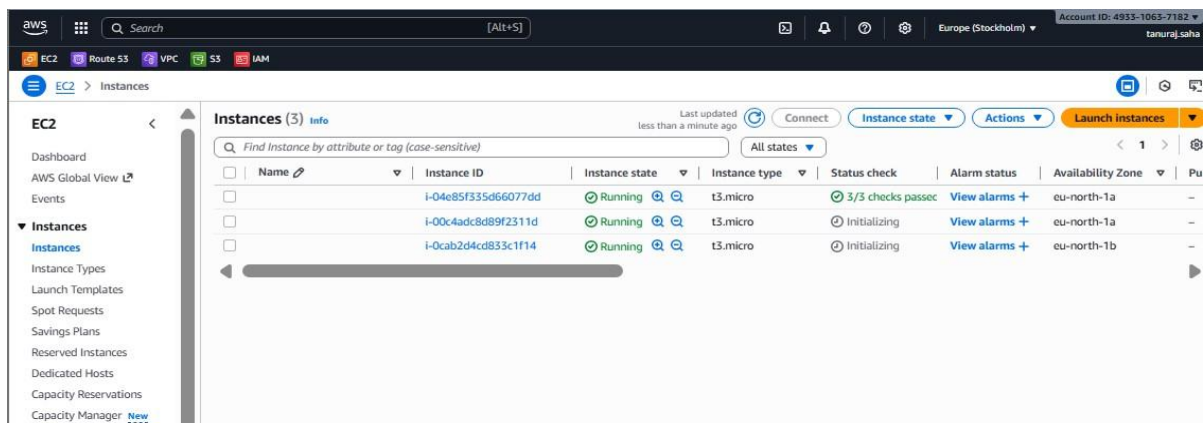
Scaling limits
Set limits on how much your desired capacity can be increased or decreased.
Min desired capacity: 1
Max desired capacity: 5
Equal or less than desired capacity Equal or greater than desired capacity

Buttons: Cancel, Update

Finally, ASG-lab-group is created and autoscaling is completed



Step 10: Now we can see 2 more servers have created automatically as we have mentioned in the Autoscaling groups before as desired capacity.

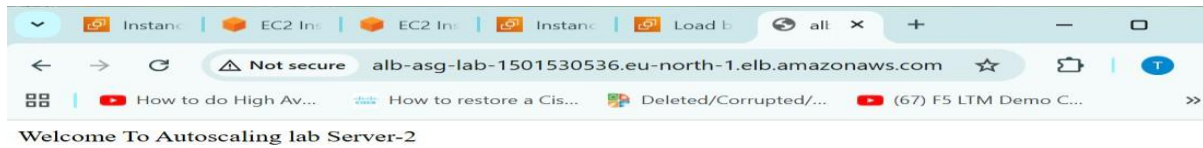


Testing: We have checked both server-1 and server 2 by curl and browsing with DNS from the ALB and found different outputs (both server-1 and server-2) every time. That's mean our traffic is load-balanced in a proper manner.

Server-1



Server-2



-----End of the Document-----

